Emotional Dynamics and Consensus Quality in Hebrew-Wikipedia Mental-Health Talk Pages

Working report (Wikimedia Israel / Dror Lin fund)

Anat Talmon Nimrod Talmon

September 2, 2026

1 Research question

Do emotional expressions in talk-page discussions (aggression, conciliation, exhaustion, contagion) predict the structure of the discussion (dropout vs. convergence) and, through it, the quality of the resulting article version—measured as *stability* (survival of a consensus version without reverts) and *correctness* (expert accuracy ratings)? Do stability and correctness agree?

2 Infrastructure

Code: `https://github.com/nimrodtalmon/Wikipediadiscussions`. Public site with homepage, corpus explorer (article → thread → conversation reader with a discussion-tree visualization), and a candidate-curation tool: `https://nimrodtalmon.github.io/Wikipediadiscussions/`. All data is drawn live from the MediaWiki API of Hebrew Wikipedia; raw data is regenerated on demand and not stored in the repository.

3 Pipeline, by module

Each module states what is implemented now and what requires joint revision. The pipeline is deliberately modular: any module can be replaced without touching the others, and the visualizations update from the rebuilt data file.

3.1 M1: Corpus definition

Currently implemented: Category crawl (depth 2) from 16 mental-health-adjacent roots on Hebrew Wikipedia: 456 categories, 7,801 articles, of which 2,957 have a talk page of at least 2 KB. The top 800 by talk size were ranked by talk-page edit counts (capped at 500 per page), yielding 834 ranked candidates. A curation interface lists them with their category path; selections export a seed list. A provisional seed of 18 articles (13 clinical core, 5 adjacent charged topics) drives all numbers below.
To revise with Anat: inclusion criteria (clinical vs. adjacent ring; politically charged articles are currently *in*), target corpus size (∼60–100 articles), and the final curated list.

3.2 M2: Data acquisition

Currently implemented: For each article: current text of the talk page and its archive subpages, plus full revision metadata (timestamps, users, sizes, comments) of both the talk pages and the article itself, via the MediaWiki API with polite rate-limiting and retries.
To revise with Anat: nothing; purely technical.

3.3 M3: Discussion segmentation

Currently implemented: Talk pages are split into threads by section headings; threads are split into comments by signature timestamps and indentation; each comment carries editor, date, indentation depth, and a reply-tree parent (nearest shallower predecessor). Known noise: unsigned fragments and inconsistent indentation conventions.
To revise with Anat: definition of a "participant turn" for the psychological analysis (e.g. merging consecutive fragments by the same editor), and how to treat unsigned comments.

3.4 M4: Emotion coding

Currently implemented: Placeholder v0: Hebrew lexicon matching for three categories—aggression, conciliation, frustration/exhaustion—plus neutral; per-comment label and intensity. On the seed corpus (2,670 comments): 175 aggressive, 564 conciliatory, 37 exhausted, 1,894 neutral. Conciliation over-fires on routine politeness; sarcasm and formal hostility are missed.
To revise with Anat: the emotion taxonomy itself (categories, definitions, granularity); coding scheme (per comment vs. per turn); the plan for LLM-based coding of rich threads; a manual coding protocol and sample size for validation (inter-rater agreement, and agreement between manual, lexicon, and LLM codings). Contagion is a derived dynamic (Module M5), not a per-comment label.

3.5 M5: Discussion dynamics

Not yet implemented. Planned per-thread measures: participation trajectories, dropout following hostile turns, convergence indicators, interaction networks, emotional contagion (autocorrelation of affect across consecutive turns).
To revise with Anat: which constructs matter psychologically, and their operational definitions.

3.6 M6: Outcome—stability

Currently implemented: Per article, from full revision history: *identity reverts* (a revision whose content hash equals an earlier revision's within a 50-edit window, i.e. the page returned exactly to a prior state), summary-based reverts as a secondary signal, edit-war episodes (≥ 3 identity reverts within 48h of one another), and version lifetimes (median and longest inter-edit gap; age of the current version). Seed-corpus headline: pedophilia is the most contested article (24% identity-revert rate, 16 war episodes); ADHD is the most edited (3,155 revisions) but among the calmest (5%).
To revise with Anat: nothing yet; window and war thresholds to be calibrated, and passage-level (rather than whole-page) stability is future work.

3.7 M7a: Outcome—quality proxy (LLM)

Currently implemented: For each rich thread, the article revision immediately after the discussion window is fetched, a passage relevant to the thread is extracted, and an LLM rates it 1–7 on accuracy, sourcing, and neutrality, with a one-sentence rationale. All 70 ratable rich threads (one lacks any post-discussion revision) were rated in-session by Claude in September 2026; scores are live in the explorer and on the article timelines. Corpus means: accuracy 4.9, sourcing 3.9, neutrality 5.0 (of 7); nine versions scored accuracy ≤ 3. The pilot already caught a substantive error: the Hebrew gender-dysphoria article's APA quotation dropped a negation, reversing the original's meaning ("gender nonconformity is *not* in itself a mental disorder"). Keys are now index-based (unique). Known v0 limits: the rated passage is located by keyword match and often falls back to the article lead, so ratings partly reflect lead quality rather than the disputed passage; and the rater is a general-purpose LLM without a jointly fixed rubric.
To revise with Anat: the rubric (dimensions, anchors) so the LLM prompt and the expert protocol are aligned.

3.8 M7b: Outcome—correctness (expert validation)

Not yet implemented; entirely joint. Expert ratings on a subsample validate (or replace) the M7a proxy: rater recruitment, rubric, unit of rating, sample size, agreement analysis.
To revise with Anat: everything; this is the clinical anchor of the project.

3.9 M8: Linking discussions to edits

Currently implemented: Time-window alignment v0: each thread is linked to the article edits between one day before its first signed comment and seven days after its last, recording edit and identity-revert counts in the window, edits whose summary cites the talk page, and *post-thread stability*: days from the thread's last comment to the next identity revert (right-censored at the present). On the seed corpus, 69 of the 71 rich threads (≥ 5 comments) have article edits in their window and 51 have reverts. Known bias: long-running threads get long windows and thus trivially many edits; rates must be normalized against the article's baseline edit rate before analysis. Content-overlap matching is future work.
To revise with Anat: nothing initially.

4 Status summary

Module	Status	Joint revision needed	Blocking
M1 corpus	candidates ready	yes: criteria + final list	M2 full run
M2 acquisition	done (seed)	no	—
M3 segmentation	done, v1	yes: turn definition	M5
M4 emotion	placeholder v0	yes: taxonomy + validation	M5 analyses
M5 dynamics	—	yes: constructs	analysis
M6 stability	done, v0	no	—
M7a quality proxy	done, v0 (70 rated)	yes: rubric	—
M7b expert validation	—	yes: protocol	analysis
M8 linking	done, v0	no	—

Table 1: Pipeline status, September 2026.

5 Seed-corpus numbers (Sept 2026)

18 articles, 474 threads, 2,670 segmented comments (1,619 signed). Activity is heavy-tailed: 71 threads have ≥ 5 signed comments; autism and ADHD dominate. Full per-article and per-thread tables are in the explorer.

6 Results: pipeline smoke test (September 2026)

This section documents a first end-to-end run whose purpose is to verify that the pipeline produces analyzable numbers—*not* to test hypotheses. Every measurement layer is v0 (lexicon emotions, time-window linking, LLM quality proxy, lead-biased passage extraction), so estimates below should be read as plumbing checks.

Method. Unit: rich threads (≥ 5 signed comments; $n = 71$; 70 with a quality rating). Predictors: shares of aggressive/conciliatory comments per thread. Outcomes: excess in-window reverts (reverts in the thread's window divided by the article's baseline revert rate times window length), post-thread stability (log days to next identity revert; right-censored), and the M7a accuracy score. Spearman correlations; Mann–Whitney for aggression-present vs. absent (uncensored only). Code: `src/analysis.py`; outputs `data/analysis.json`.

Test	ρ	p	n
Aggression share vs. excess in-window reverts	−0.10	.42	71
Aggression share vs. post-stability	+0.16	.20	71
Conciliation share vs. post-stability	−0.15	.20	71
Aggression share vs. accuracy	−0.20	.10	70
Post-stability vs. accuracy	−0.04	.77	70
Number of editors vs. post-stability	−0.36	.002	71
Article sentiment balance vs. revert rate	+0.14	.58	18

Table 2: Spearman correlations, v0 smoke test. Post-stability is $\log_{10}(\text{days}+1)$.

Findings. Three observations worth carrying forward. (1) The only robust association is structural: threads with more editors precede less stable versions ($\rho = -.36$)—plausibly confounded by article activity, but a sensible sanity check that the linking layer carries signal. (2) Stability and accuracy are uncorrelated ($\rho = -.04$): if this survives better measurement, it is the paper's central provocation—Wikipedia can be stably wrong. (3) The emotion associations are directionally sensible (aggression weakly negative for accuracy) but nowhere near significance, as expected given lexicon-level measurement; this is the strongest argument for prioritizing the M4 taxonomy work.

What must improve before any claim. M4 coding quality (LLM + validation); M8 passage-level linking instead of time windows; survival analysis honoring censoring instead of log-days; article-level clustering (autism and ADHD dominate); and M7a rubric alignment with M7b.

7 Discussion